AIAC-ASSIGNMENT-11.2

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Task Description #1 – Stack Implementation

Task: Use AI to generate a Stack class with push, pop, peek, and is_empty methods.

Sample Input Code:

class Stack:

pass

Expected Output:

• A functional stack implementation with all required methods and docstrings.

```
# generate a python code to demonstrate stack and its operations
class Stack:
    def __init__(self):
        self.items = []

    def is_empty(self):
        return len(self.items) == 0

    def push(self, item):
        self.items.append(item)

    def pop(self):
        if not self.is_empty():
            return self.items.pop()
        else:
            raise IndexError("pop from empty stack")

    def peek(self):
        if not self.is_empty():
            return self.items[-1]
        else:
            raise IndexError("peek from empty stack")

    def size(self):
        return len(self.items)
```

```
# example
stack = Stack()
stack.push(1)
stack.push(2)
stack.push(3)
print(stack.peek())   # Output: 3
print(stack.size())   # Output: 3
print(stack.pop())    # Output: 3
print(stack.pop())    # Output: 2
print(stack.is_empty())   # Output: False
```

Task Description #2 – Queue Implementation

Task: Use AI to implement a Queue using Python lists.

Sample Input Code:

class Queue:

pass

Expected Output:

• FIFO-based queue class with enqueue, dequeue, peek, and size

methods.

```
# generate a python code to demonstrate queue and its operations
class Queue:
    def __init__(self):
        self.items = []

    def is_empty(self):
        return len(self.items) == 0

    def enqueue(self, item):
        self.items.append(item)

    def dequeue(self):
        if not self.is_empty():
            return self.items.pop(0)
        else:
            raise IndexError("dequeue from empty queue")

    def peek(self):
        if not self.is_empty():
            return self.items[0]
        else:
            raise IndexError("peek from empty queue")
```

```
    def size(self):
        return len(self.items)
#example
queue = Queue()
queue.enqueue(1)
queue.enqueue(2)
queue.enqueue(3)
print(queue.peek())   # Output: 1
print(queue.size())   # Output: 3
print(queue.dequeue())    # Output: 1
print(queue.dequeue())    # Output: 2
print(queue.is_empty())   # Output: False
```

Task Description #3 – Linked List

Task: Use AI to generate a Singly Linked List with insert and display

methods.

Sample Input Code:

class Node:

pass

class LinkedList:

pass

Expected Output:

• A working linked list implementation with clear method

documentation.

```
# generate a python code to demonstrate linked list and its operations
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
class LinkedList:
    def __init__(self):
        self.head = None

    def is_empty(self):
        return self.head is None

    def append(self, data):
        new_node = Node(data)
        if self.is_empty():
```

```
            self.head = new_node
            return
        last_node = self.head
        while last_node.next:
            last_node = last_node.next
        last_node.next = new_node

    def delete(self, key):
        current_node = self.head
        previous_node = None
        while current_node and current_node.data != key:
            previous_node = current_node
            current_node = current_node.next
        if current_node is None:
            return  # Key not found
        if previous_node is None:
            self.head = current_node.next  # Deleting the head node
        else:
            previous_node.next = current_node.next  # Bypass the node to delete

    def search(self, key):
        current_node = self.head
        while current_node:
            if current_node.data == key:
                return True
            current_node = current_node.next
        return False

    def size(self):
        count = 0
        current_node = self.head
        while current_node:
            count += 1
            current_node = current_node.next
        return count
# example
linked_list = LinkedList()
linked_list.append(1)
linked_list.append(2)
linked_list.append(3)
print(linked_list.search(2))  # Output: True
print(linked_list.size())  # Output: 3
linked_list.delete(2)
print(linked_list.search(2))  # Output: False
print(linked_list.size())  # Output: 2
```

Task Description #4 – Binary Search Tree (BST)

Task: Use AI to create a BST with insert and in-order traversal methods.

Sample Input Code:

class BST:

pass

Expected Output:

- BST implementation with recursive insert and traversal methods.

```
# generate a python code to demonstrate binary search tree and its operations
class TreeNode:
    def __init__(self, key):
        self.left = None
        self.right = None
        self.val = key
class BinarySearchTree:
    def __init__(self):
        self.root = None

    def insert(self, key):
        if self.root is None:
            self.root = TreeNode(key)
        else:
            self._insert_recursively(self.root, key)

    def _insert_recursively(self, node, key):
        if key < node.val:
            if node.left is None:
                node.left = TreeNode(key)
            else:
                self._insert_recursively(node.left, key)
        else:
            if node.right is None:
                node.right = TreeNode(key)
            else:
                self._insert_recursively(node.right, key)

    def search(self, key):
        return self._search_recursively(self.root, key)

    def _search_recursively(self, node, key):
        if node is None:
            return False
        if node.val == key:
            return True
        elif key < node.val:
            return self._search_recursively(node.left, key)
```

```
        else:
            return self._search_recursively(node.right, key)

    def inorder_traversal(self):
        return self._inorder_traversal_recursively(self.root)

    def _inorder_traversal_recursively(self, node):
        res = []
        if node:
            res = self._inorder_traversal_recursively(node.left)
            res.append(node.val)
            res = res + self._inorder_traversal_recursively(node.right)
        return res
# example
bst = BinarySearchTree()
bst.insert(5)
bst.insert(3)
bst.insert(7)
print(bst.search(3))   # Output: True
print(bst.search(4))   # Output: False
print(bst.inorder_traversal())   # Output: [3, 5, 7]
```

Task Description #5 – Hash Table

Task: Use AI to implement a hash table with basic insert, search, and

delete methods.

Sample Input Code:

class HashTable:

pass

Expected Output:

- Collision handling using chaining, with well-commented methods.

```
# generate a python code to demonstrate hash table and its operations
class HashTable:
    def __init__(self):
        self.table = {}

    def insert(self, key, value):
        self.table[key] = value

    def delete(self, key):
        if key in self.table:
            del self.table[key]
```

```
    def search(self, key):
        return self.table.get(key, None)

    def size(self):
        return len(self.table)
# example
hash_table = HashTable()
hash_table.insert("name", "Alice")
hash_table.insert("age", 30)
print(hash_table.search("name"))  # Output: Alice
print(hash_table.size())  # Output: 2
hash_table.delete("age")
print(hash_table.search("age"))  # Output: None
print(hash_table.size())  # Output: 1
```

Task Description #6 – Graph Representation

Task: Use AI to implement a graph using an adjacency list.

Sample Input Code:

class Graph:

pass

Expected Output:

• Graph with methods to add vertices, add edges, and display

connections.

```
# generate a python code to demonstrate graph and its operations
class Graph:
    def __init__(self):
        self.graph = {}

    def add_edge(self, u, v):
        if u not in self.graph:
            self.graph[u] = []
        if v not in self.graph:
            self.graph[v] = []
        self.graph[u].append(v)
        self.graph[v].append(u)  # For undirected graph

    def bfs(self, start):
        visited = set()
        queue = [start]
        while queue:
```

```
            vertex = queue.pop(0)
            if vertex not in visited:
                print(vertex, end=' ')
                visited.add(vertex)
                queue.extend(set(self.graph[vertex]) - visited)

    def dfs(self, start, visited=None):
        if visited is None:
            visited = set()
        if start not in visited:
            print(start, end=' ')
            visited.add(start)
            for neighbor in self.graph[start]:
                self.dfs(neighbor, visited)

# example
graph = Graph()
graph.add_edge(0, 1)
graph.add_edge(0, 2)
graph.add_edge(1, 2)
graph.add_edge(2, 0)
graph.add_edge(2, 3)
print("Breadth-First Search starting from vertex 2:")
graph.bfs(2)   # Output: 2 0 3 1
print("\nDepth-First Search starting from vertex 2:")
graph.dfs(2)   # Output: 2 0 1 3
```

Task Description #7 – Priority Queue

Task: Use AI to implement a priority queue using Python's heapq

module.

Sample Input Code:

class PriorityQueue:

pass

Expected Output:

• Implementation with enqueue (priority), dequeue (highest priority),

and display methods.

```
# generate a python code to demonstrate priority queue and its operations
import heapq
class PriorityQueue:
    def __init__(self):
        self.elements = []
```

```
    def is_empty(self):
        return not self.elements

    def put(self, item, priority):
        heapq.heappush(self.elements, (priority, item))

    def get(self):
        return heapq.heappop(self.elements)[1]

    def peek(self):
        return self.elements[0][1] if self.elements else None

    def size(self):
        return len(self.elements)
# example
priority_queue = PriorityQueue()
priority_queue.put("task1", 2)
priority_queue.put("task2", 1)
print(priority_queue.peek())   # Output: task2
print(priority_queue.size())   # Output: 2
print(priority_queue.get())    # Output: task2
print(priority_queue.get())    # Output: task1
```

Task Description #8 – Deque

Task: Use AI to implement a double-ended queue using

collections.deque.

Sample Input Code:

class DequeDS:

pass

Expected Output:

- Insert and remove from both ends with docstrings.

```
# generate a python code to demonstrate deque and its operations
class Deque:
    def __init__(self):
        self.items = []

    def is_empty(self):
        return len(self.items) == 0

    def add_front(self, item):
        self.items.insert(0, item)
```

```
    def add_rear(self, item):
        self.items.append(item)

    def remove_front(self):
        if not self.is_empty():
            return self.items.pop(0)
        else:
            raise IndexError("remove from empty deque")

    def remove_rear(self):
        if not self.is_empty():
            return self.items.pop()
        else:
            raise IndexError("remove from empty deque")

    def peek_front(self):
        if not self.is_empty():
            return self.items[0]
        else:
            raise IndexError("peek from empty deque")

    def peek_rear(self):
        if not self.is_empty():
            return self.items[-1]
        else:
            raise IndexError("peek from empty deque")

    def size(self):
        return len(self.items)
# example
deque = Deque()
deque.add_rear(1)
deque.add_rear(2)
deque.add_front(0)
print(deque.peek_front())   # Output: 0
print(deque.peek_rear())    # Output: 2
print(deque.size())         # Output: 3
print(deque.remove_front()) # Output: 0
print(deque.remove_rear())  # Output: 2
print(deque.is_empty())     # Output: False
```

Task Description #9 Real-Time Application Challenge – Choose the Right Data Structure

Scenario:

Your college wants to develop a Campus Resource Management System that handles:

1. Student Attendance Tracking – Daily log of students entering/exiting the campus.
2. Event Registration System – Manage participants in events with quick search and removal.
3. Library Book Borrowing – Keep track of available books and their due dates.
4. Bus Scheduling System – Maintain bus routes and stop connections.
5. Cafeteria Order Queue – Serve students in the order they arrive.

Student Task:

- For each feature, select the most appropriate data structure from the list below:
  - o Stack
  - o Queue
  - o Priority Queue
  - o Linked List
  - o Binary Search Tree (BST)
  - o Graph
  - o Hash Table
  - o Deque
- Justify your choice in 2–3 sentences per feature.

Feature	Data Structure	Justification
Student Attendance Tracking	**Linked List**	Attendance records are continuously added every day. A linked list allows easy insertion of new records without shifting existing data.
Event Registration System	**Binary Search Tree (BST)**	Participants can be stored based on registration ID or name. BST allows efficient searching, insertion, and removal of participants.

Feature	Data Structure	Justification
Library Book Borrowing	**Hash Table**	Books can be accessed using a unique book ID as a key. Hash tables allow very fast lookup and update of book availability.
Bus Scheduling System	**Graph**	Bus routes connect multiple stops forming a network. Graphs represent these connections efficiently and allow route traversal.
Cafeteria Order Queue	**Queue**	Orders are served in the order students arrive. Queue follows the FIFO principle which perfectly matches this process.

• Implement one selected feature as a working Python program with

AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature

with comments and docstrings.

```
# generate a python code for bus scheduling system using graph data structure
class Graph:
    def __init__(self):
        self.graph = {}
    def add_edge(self, u, v):
        if u not in self.graph:
            self.graph[u] = []
        if v not in self.graph:
            self.graph[v] = []
        self.graph[u].append(v)
        self.graph[v].append(u)   # For undirected graph
    def bfs(self, start):
        visited = set()
        queue = [start]
        while queue:
            vertex = queue.pop(0)
            if vertex not in visited:
                print(vertex, end=' ')
                visited.add(vertex)
                queue.extend(set(self.graph[vertex]) - visited)
    def dfs(self, start, visited=None):
        if visited is None:
```

```
            visited = set()
        if start not in visited:
            print(start, end=' ')
            visited.add(start)
            for neighbor in self.graph[start]:
                self.dfs(neighbor, visited)
# example
bus_schedule = Graph()
bus_schedule.add_edge("Stop A", "Stop B")
bus_schedule.add_edge("Stop A", "Stop C")
bus_schedule.add_edge("Stop B", "Stop D")
bus_schedule.add_edge("Stop C", "Stop D")
print("Bus route from Stop A:")
bus_schedule.bfs("Stop A")   # Output: Stop A Stop B Stop C Stop D
print("\nBus route from Stop A (DFS):")
bus_schedule.dfs("Stop A")   # Output: Stop A Stop B Stop D Stop C
```

Task Description #10: Smart Hospital Management System – Data Structure Selection

A hospital wants to develop a Smart Hospital Management System that handles:

1. Patient Check-In System – Patients are registered and treated in order of arrival.

2. Emergency Case Handling – Critical patients must be treated first.

3. Medical Records Storage – Fast retrieval of patient details using ID.

4. Doctor Appointment Scheduling – Appointments sorted by time.

5. Hospital Room Navigation – Represent connections between wards and rooms.

Student Task

• For each feature, select the most appropriate data structure from the list below:

o Stack

o Queue

o Priority Queue

o Linked List

o Binary Search Tree (BST)

o Graph

o Hash Table

o Deque

• Justify your choice in 2–3 sentences per feature.

Feature	Data Structure	Justification
Patient Check-In System	**Queue**	Patients are treated in the order they arrive. Queue ensures FIFO processing of patient entries.
Emergency Case Handling	**Priority Queue**	Critical patients must be treated before others regardless of arrival time. Priority queue processes elements based on urgency level.
Medical Records Storage	**Hash Table**	Each patient record can be accessed using a unique patient ID. Hash tables provide very fast retrieval of records.
Doctor Appointment Scheduling	**Binary Search Tree (BST)**	Appointments can be organized based on time. BST allows quick insertion and ordered traversal of time slots.
Hospital Room Navigation	**Graph**	Hospital rooms and wards are interconnected like nodes and edges. Graphs help represent and navigate these connections.

• Implement one selected feature as a working Python program with

AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature

with comments and docstrings.

```python
#generate a python code for hash table implementation for medical records storage
class HashTable:
    def __init__(self, size=100):
        self.size = size
        self.table = [None] * self.size

    def _hash(self, key):
        return hash(key) % self.size

    def insert(self, key, value):
        index = self._hash(key)
        if self.table[index] is None:
            self.table[index] = [(key, value)]
        else:
            for i, (k, v) in enumerate(self.table[index]):
                if k == key:
                    self.table[index][i] = (key, value)  # Update existing key
                    return
            self.table[index].append((key, value))  # Add new key-value pair

    def get(self, key):
        index = self._hash(key)
        if self.table[index] is not None:
            for k, v in self.table[index]:
                if k == key:
                    return v
        return None  # Key not found

    def delete(self, key):
        index = self._hash(key)
        if self.table[index] is not None:
            for i, (k, v) in enumerate(self.table[index]):
                if k == key:
                    del self.table[index][i]  # Remove the key-value pair
                    return True
        return False  # Key not found
```

```python
        return False  # Key not found
# Example usage
if __name__ == "__main__":
    medical_records = HashTable()
    medical_records.insert("patient_001", {"name": "John Doe", "age": 30, "condition": "Flu"})
    medical_records.insert("patient_002", {"name": "Jane Smith", "age": 25, "condition": "Cold"})

    print(medical_records.get("patient_001"))  # Output: {'name': 'John Doe', 'age': 30, 'condition': 'Flu'}
    print(medical_records.get("patient_002"))  # Output: {'name': 'Jane Smith', 'age': 25, 'condition': 'Cold'}

    medical_records.delete("patient_001")
    print(medical_records.get("patient_001"))  # Output: None
```

```
{'name': 'John Doe', 'age': 30, 'condition': 'Flu'}
{'name': 'Jane Smith', 'age': 25, 'condition': 'Cold'}
None
```

Task Description #11: Smart City Traffic Control System

A city plans a Smart Traffic Management System that includes:

1. Traffic Signal Queue – Vehicles waiting at signals.

2. Emergency Vehicle Priority Handling – Ambulances and fire trucks prioritized.

3. Vehicle Registration Lookup – Instant access to vehicle details.

4. Road Network Mapping – Roads and intersections connected logically.

5. Parking Slot Availability – Track available and occupied slots.

Student Task

• For each feature, select the most appropriate data structure from the list below:

o Stack

o Queue

o Priority Queue

o Linked List

o Binary Search Tree (BST)

o Graph

o Hash Table

o Deque

• Justify your choice in 2–3 sentences per feature.

Feature	Data Structure	Justification
Traffic Signal Queue	**Queue**	Vehicles move through the signal in the order they arrive. Queue naturally models this FIFO process.
Emergency Vehicle Priority	**Priority Queue**	Emergency vehicles must pass first regardless of arrival order. Priority queue ensures high-priority vehicles are processed first.
Vehicle Registration Lookup	**Hash Table**	Vehicle details can be stored using registration numbers as keys. Hash tables allow instant retrieval of vehicle information.

Feature	Data Structure	Justification
Road Network Mapping	Graph	Roads connect intersections forming a network structure. Graphs efficiently represent these connections for navigation.
Parking Slot Availability	Hash Table	Each slot can be mapped using a unique slot number. Hash tables allow quick updates for available or occupied slots.

• Implement one selected feature as a working Python program with

AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature

with comments and docstrings.

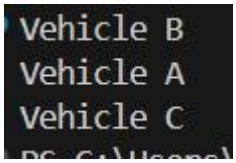
```python
#generate a python code for priority queue implementation for vehicle priority handling
class PriorityQueue:
    def __init__(self):
        self.elements = []

    def is_empty(self):
        return not self.elements

    def put(self, item, priority):
        self.elements.append((priority, item))
        self.elements.sort(key=lambda x: x[0])  # Sort by priority

    def get(self):
        if not self.is_empty():
            return self.elements.pop(0)[1]  # Return the item with the highest priority
        raise IndexError("Priority Queue is empty")
# Example usage
if __name__ == "__main__":
    pq = PriorityQueue()
    pq.put("Vehicle A", 2)  # Lower number means higher priority
    pq.put("Vehicle B", 1)
    pq.put("Vehicle C", 3)

    print(pq.get())  # Output: Vehicle B (highest priority)
    print(pq.get())  # Output: Vehicle A
    print(pq.get())  # Output: Vehicle C
    # print(pq.get())  # This will raise an error since the queue is empty
```



Task Description #12: Smart E-Commerce Platform – Data Structure Challenge

An e-commerce company wants to build a Smart Online Shopping System with:

1. Shopping Cart Management – Add and remove products dynamically.
2. Order Processing System – Orders processed in the order they are placed.
3. Top-Selling Products Tracker – Products ranked by sales count.
4. Product Search Engine – Fast lookup of products using product ID.
5. Delivery Route Planning – Connect warehouses and delivery locations.

Student Task

- For each feature, select the most appropriate data structure from the list below:
  - o Stack
  - o Queue
  - o Priority Queue
  - o Linked List
  - o Binary Search Tree (BST)
  - o Graph
  - o Hash Table
  - o Deque
- Justify your choice in 2–3 sentences per feature.

Feature	Data Structure	Justification

Feature	Data Structure	Justification
Shopping Cart Management	**Linked List**	Products can be dynamically added or removed from the cart. Linked lists allow flexible insertion and deletion operations.
Order Processing System	**Queue**	Orders must be processed in the order they are placed. Queue ensures FIFO order processing.
Top-Selling Products Tracker	**Priority Queue**	Products can be ranked by sales count. Priority queue allows quick access to highest-selling products.
Product Search Engine	**Hash Table**	Products can be retrieved instantly using product IDs. Hash tables provide constant-time lookup.
Delivery Route Planning	**Graph**	Warehouses and delivery locations form a connected network. Graphs help compute routes between locations.

• Implement one selected feature as a working Python program with

AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature

with comments and docstrings.

```python
#generate a python code for graph implementation for delivery route planning
class Graph:
    def __init__(self):
        self.graph = {}

    def add_edge(self, u, v, weight=1):
        if u not in self.graph:
            self.graph[u] = []
        if v not in self.graph:
            self.graph[v] = []
        self.graph[u].append((v, weight))
        self.graph[v].append((u, weight))  # For undirected graph

    def dijkstra(self, start):
        import heapq
        queue = [(0, start)]
        distances = {node: float('inf') for node in self.graph}
        distances[start] = 0
        while queue:
            current_distance, current_node = heapq.heappop(queue)
            if current_distance > distances[current_node]:
                continue
            for neighbor, weight in self.graph[current_node]:
                distance = current_distance + weight
                if distance < distances[neighbor]:
                    distances[neighbor] = distance
                    heapq.heappush(queue, (distance, neighbor))
        return distances
# Example usage
if __name__ == "__main__":
    g = Graph()
    g.add_edge("Warehouse", "Customer A", 5)
    g.add_edge("Warehouse", "Customer B", 10)
    g.add_edge("Customer A", "Customer B", 2)

    distances = g.dijkstra("Warehouse")
    print(distances)  # Output: {'Warehouse': 0, 'Customer A': 5, 'Customer B': 7}
```

```
{'Warehouse': 0, 'Customer A': 5, 'Customer B': 7}
```

Task Description #14: Smart Banking Management System – Data Structure Challenge

A bank plans to develop a Smart Banking System that includes:

1. Customer Service Token System – Customers are served in order of arrival.

2. Loan Approval Priority Handling – High-value or urgent loans processed first.

3. Account Information Lookup – Fast retrieval of customer details

using Account Number.

4. Transaction History Management – Maintain chronological transaction records.

5. ATM Network Connectivity – Represent connections between ATMs across cities.

Student Task:

• For each feature, select the most appropriate data structure from the list below:

o Stack

o Queue

o Priority Queue

o Linked List

o Binary Search Tree (BST)

o Graph

o Hash Table

o Deque

• Justify your choice in 2–3 sentences per feature.

Feature	Data Structure	Justification
Customer Service Token System	Queue	Customers are served in the order they arrive. Queue follows FIFO, matching the service process.
Loan Approval Priority	Priority Queue	High-value or urgent loans must be processed first. Priority queue processes items based on priority levels.
Account Information Lookup	Hash Table	Accounts are uniquely identified by account numbers. Hash tables allow extremely fast data retrieval.
Transaction History Management	Linked List	Transactions occur sequentially over time. Linked lists efficiently store and append chronological records.
ATM Network Connectivity	Graph	ATMs across cities form a network of connections. Graphs represent paths between ATMs efficiently.

• Implement one selected feature as a working Python program

with AI-assisted code generation.

Expected Output:

• A table mapping feature → chosen data structure → justification.

• A functional Python program implementing the chosen feature

with comments and docstrings.

```python
#generate a python code for queue  implementation for Customer Service Token System
class Queue:
    def __init__(self):
        self.elements = []

    def is_empty(self):
        return not self.elements

    def enqueue(self, item):
        self.elements.append(item)

    def dequeue(self):
        if not self.is_empty():
            return self.elements.pop(0)  # Remove and return the first item
        raise IndexError("Queue is empty")
# Example usage
if __name__ == "__main__":
    customer_queue = Queue()
    customer_queue.enqueue("Customer 1")
    customer_queue.enqueue("Customer 2")
    customer_queue.enqueue("Customer 3")

    print(customer_queue.dequeue())  # Output: Customer 1
    print(customer_queue.dequeue())  # Output: Customer 2
    print(customer_queue.dequeue())  # Output: Customer 3
    # print(customer_queue.dequeue())  # This will raise an error since the queue is empty
```

```
Customer 1
Customer 2
Customer 3
```

Task Description #15: Online Learning Management System – Data

Structure Selection

An online education platform wants to develop a Learning Management

System that handles:

1. Student Login Authentication – Quick validation of user

credentials.

2. Assignment Submission Queue – Submissions processed in order

of upload.

3. Top Performer Ranking – Rank students based on scores.

4. Course Prerequisite Structure – Represent subject dependencies.

5. Undo Feature in Online Editor – Reverse recent actions.

Student Task:

• Select the most appropriate data structure for each feature from

the given list.

• Justify your selection in 2–3 sentences per feature.

Feature	Data Structure	Justification
Student Login Authentication	Hash Table	User credentials can be mapped with usernames as keys. Hash tables allow instant authentication lookup.
Assignment Submission Queue	Queue	Assignments are processed in the order they are submitted. Queue ensures fair FIFO processing.
Top Performer Ranking	Priority Queue	Students with higher scores should appear first. Priority queue automatically orders students by score.
Course Prerequisite Structure	Graph	Courses depend on other prerequisite courses. Graphs represent these dependency relationships clearly.
Undo Feature in Editor	Stack	The most recent action must be undone first. Stack follows LIFO, which matches undo operations.

• Implement one selected feature in Python using AI-assisted code

generation.

Expected Output:

• Feature → Data Structure → Justification table.

• Functional Python program with proper documentation.

```python
#generate a python code for undo feature in online text editor using stack data structure
class Stack:
    def __init__(self):
        self.elements = []

    def is_empty(self):
        return not self.elements

    def push(self, item):
        self.elements.append(item)

    def pop(self):
        if not self.is_empty():
            return self.elements.pop()  # Remove and return the last item
        raise IndexError("Stack is empty")
# Example usage
if __name__ == "__main__":
    undo_stack = Stack()
    undo_stack.push("Type 'Hello'")
    undo_stack.push("Type 'World'")
    undo_stack.push("Delete 'World'")

    print(undo_stack.pop())  # Output: Delete 'World'
    print(undo_stack.pop())  # Output: Type 'World'
    print(undo_stack.pop())  # Output: Type 'Hello'
    # print(undo_stack.pop())  # This will raise an error since the stack is empty
```

```
Delete 'World'
Type 'World'
Type 'Hello'
```